Separating Rate from Composition: A Zero-Loss Decomposition of Electoral Change

Donald Moratz

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Abstract

elections

why not just important?

Voters' group identities structure electoral behavior, making accurate decomposition of how electoral coalitions shift between ~~cycles~~ ~~increasingly~~ important. Yet existing methods — whether regression-based or ~~the most~~ more recent non-linear approaches — introduce systematic error, silently misassigning votes across components and producing residuals exceeding one million votes. I introduce the Rate, Composition, and Volume Decomposition (RCVD), a zero-loss framework that partitions total vote change into three components: changes in group loyalty (rate), group size (composition), and overall participation (volume). Through sequential rather than simultaneous calculation, RCVD eliminates residual error while preserving substantive interpretability. Applied to U.S. presidential elections from 2016 to 2024, RCVD reveals that the 2024 result was driven by a decisive decoupling of rate and composition effects among Hispanic voters. Demographic growth favored Democrats compositionally; rightward shifts in Hispanic vote margins neutralized that advantage entirely.

Maybe put this in percentage terms?

1 Introduction

Voters' identification with social groups structures their electoral behavior (Campbell, 1960). To the extent people vote according to social group, understanding how

membership

the composition of these groups shifts over time is key to understanding aggregate outcomes (Axelrod, 1972). This is particularly relevant in an era of increasing polarization, where electoral success requires building and maintaining broad coalitions across identity groups (Sides et al., 2019). Yet the methods political scientists use to decompose electoral change between cycles introduce systematic error in that attribution — silently misassigning hundreds of thousands of votes and obscuring the true drivers of electoral change.

The challenge is inherent to the problem's structure. Current approaches — whether regression-based (Hill et al., 2021) or the recent and ~~important~~ linear difference framework of Marble et al. (2024) — face a fundamental tension: separating shifts in group voting preferences from shifts in the demographic composition of the electorate requires either accepting uninterpretable interaction terms or imposing restrictive functional form assumptions. When scholars omit these interaction terms to maintain interpretability, as is standard practice, the resulting estimates systematically misattribute vote changes. In close elections, residuals of this magnitude can reverse substantive conclusions about which factors drove the outcome. Marble et al. (2024) represent the state of the art, but ~~their~~ its decomposition yields a residual of 1.2 million votes — votes that are real but mathematically unassigned to any component.

This paper introduces the Rate, Composition, and Volume Decomposition (RCVD), which resolves this problem through sequential rather than simultaneous calculation. Three terms do all the work: *Rate*, the two-party vote margin within a demographic group; *Composition*, that group's share of the total electorate; and *Volume*, the total number of voters casting ballots. Each group's contribution to a candidate's vote margin is the product of these three quantities. By updating each component sequentially — first allowing rate to shift, then composition, then volume — RCVD accounts for the interactions between components without requiring separate interaction terms. The result is an exact decomposition with zero residual error, and each component retains a clear substantive interpretation.

Applied to U.S. presidential elections from 2016 to 2024, RCVD reveals that the

I avoid such adjectives — but you might say widely employed and cite Freya et al.

election

I am not sure this follows — or that you need this claim

you sound 1 million in the abstract; why not state in percentage terms?

This sentence is vital, so consider breaking it up for clarity.

The interactions are so central — can you provide 1-2 sentences of intuition about how they arise? Why they are needed?

2024 result was driven by a decisive decoupling of rate and composition effects among Hispanic voters. While demographic growth continued to favor Democrats compositionally, rightward shifts in Hispanic vote margins neutralized that advantage entirely — a pattern invisible to existing methods. Cleanly separating rate from composition is what makes the decoupling visible.

The article proceeds as follows. Section 2 reviews existing decomposition methods and develops the RCVD framework, demonstrating why standard approaches generate residual error. Section 3 presents worked examples with both hypothetical and real data, validating RCVD through controlled scenarios. Section 4 applies RCVD to U.S. presidential elections from 2016 to 2024, revealing how the method exposes data quality issues and captures Hispanic voting dynamics that existing approaches miss. Section 5 concludes.

2 Historical Approaches and the RCVD Framework

Measuring how electoral coalitions shift across elections requires distinguishing two fundamentally different dynamics: changes in how groups vote, and changes in who comprises the electorate. Political scientists have developed a range of approaches to this problem, each making different tradeoffs between interpretability and precision.

People typically use the singular for papers

Contemporary scholarship has increasingly emphasized the role of identity-based coalitions in shaping electoral outcomes. Achen and Bartels (2016) argue that political science needs to move away from an exclusive focus on individuals and toward greater attention to identity groups. Sides et al. (2019) demonstrate that the 2016 Trump campaign succeeded in part through reshaping party coalitions around identity politics. In this context, Lemi (2021) demonstrate that building electoral coalitions based on group membership has become a central campaign strategy — which means that accurately separating rate from composition effects is not merely a technical exercise but a substantive one.

I am not sure the "increasingly" is necessary or accurate. See Silva Kim's Political Behavior article.

Fraga et al.?

Recent scholarship has developed various approaches to decompose vote changes into their constituent parts. Hill et al. (2021) uses a regression-based approach to

attribute vote share changes to either composition or conversion, while Zingher (2019) measures how many votes each social group contributes to party coalitions across elections, distinguishing demographic from behavioral forces driving those changes. However, regression-based approaches impose strict linear specifications on the functional form of the relationship, and misspecification can introduce significant bias even when the relationship is otherwise well-specified (Wooldridge, 2010). As Engelhardt (2019) points out, shifts in the demographic composition of the electorate and shifts in vote choice within groups can mask each other, making regression coefficients difficult to interpret and obscuring the true drivers of electoral change.

But this is a generic critique of regression, no?

what specifically is non-linear? e.g. why does regression fail in this application

I think you need a sentence or two more to develop this critique — why can't regression disentangle these different factors?

Other analyses have employed simpler extrapolation methods. Fraga, McElwee, et al. (2021) examines how racial differences in turnout affected the 2016 election by applying 2012 turnout rates to construct counterfactual vote shares, isolating the contribution of differential turnout from shifts in vote choice. Carmines et al. (2016) decompose each group's contribution to the party coalitions into size, turnout, and loyalty components, using this framework to show how populist voters' distinctive profile across these dimensions helped deliver the 2016 outcome to Trump. These studies illuminate specific aspects of how rate and composition shifts drive electoral change, but neither provides a framework that accounts for how those components interact when they move simultaneously.

→ Another sentence here would help – what do such interactions look like?

The importance of accurate decomposition extends beyond single elections to forecasting. Calvo et al. (2024) demonstrate that even perfect knowledge of demographic shifts provides insufficient information to guarantee accurate forecasts — cleanly separating shifts in group size from shifts in how groups vote is essential to understanding why forecasts succeed or fail. Grimmer, Knox, et al. (2024) argue that properly evaluating forecasts requires understanding how shifts in group composition and vote choice interact across multiple election cycles. Together these contributions establish that the interaction between rate and composition effects is not a methodological curiosity but a substantive problem with direct consequences for how well we understand and anticipate electoral outcomes.

You invoke these interactions again and again but don't seem to explain them. Because you haven't yet formally stated the problem, I have a hard time following this talk of interactions

The most comprehensive framework to date is Marble et al. (2024), who reject the linearity assumption of earlier regression-based approaches and adopt a non-linear specification that directly models how each group's vote margin (Rate) and electoral share (Composition) changes between elections. Explicitly, rate is a net measure: if 60% of a group votes Republican and 40% votes Democrat, the margin rate is +20% (or +0.20), meaning that group contributes a net positive margin to Republicans. Marble et al. (2024) capture each group's contribution to total vote change as:

$$\begin{aligned}\text{Diff Net}_{t,t-1}(x) = {} & [\text{Vote Share}_t(x, \text{Republican}) - \text{Vote Share}_t(x, \text{Democrat})] \\ & \times \text{Turnout}_t(x) \times \text{Group Size}_t(x) \\ & - [\text{Vote Share}_{t-1}(x, \text{Republican}) - \text{Vote Share}_{t-1}(x, \text{Democrat})] \\ & \times \text{Turnout}_{t-1}(x) \times \text{Group Size}_{t-1}(x)\end{aligned} \tag{1}$$

Constraints? Do what sums to 1?

Where $\text{Turnout}_t(x) \times \text{Group Size}_t(x)$ represents the group's share of total votes cast. Here x indexes the group of interest and t indexes elections rather than years.[1] To separate the rate effect from the composition effect, they calculate each component independently: first fixing group composition (in their terminology, the combined turnout and group size term) at the baseline election and measuring the impact of changed vote shares, then fixing vote shares at the baseline and measuring the impact of changed composition. The two components are then summed to estimate each group's total contribution to electoral change. Fraga, Velez, et al. (2024) apply this same decomposition framework to Latino subgroups, finding that Trump's 2020 gains reflected shifts in vote choice concentrated among subgroups already predisposed toward Republicans — rather than a uniform shift across Latino voters broadly.

However, what I term the Linear Difference Approach (LDA) faces two key challenges. First, it conflates Composition and Volume — treating them as a single combined term rather than separating them. In RCVD, Composition refers strictly to the relative share of the electorate each group comprises (for example, 60 percent White,

Is this your term? Since LDA already means "Latent Dirichlet Allocation" is there another possible name?

[1] i.e. the 2016 U.S. presidential election might be $t = 0$ and 2020 might be $t = 1$.

40 percent non-White), while Volume is the total number of voters casting ballots. The LDA combines these into a single term — group size multiplied by turnout rate — which captures whether a group contributed more or fewer votes overall, but cannot distinguish whether that change came from the group becoming a larger share of the electorate or from the electorate simply expanding. Consider turnout in the 2008 presidential election: turnout reached a 40-year high (Woolley and Peters, 2021), but the aggregate increase masked important differential shifts. Turnout among Black voters rose by nearly 5 percent while turnout among White voters declined by just over 1 percent (Lopez and Taylor, 2009). The LDA would capture positive effects for Black voters and negative effects for White voters, but by conflating group share with total electorate size, it understates the compositional shift. Black voters became a meaningfully larger share of the electorate, not just a group that turned out more.

But why can't we just normalize to percentages — what's the downside? Maybe clarify which quantities of interest this affects and which is does not. Does this simply mean the Markle et al. approach can't capture some quantities (like absolute numbers of voters), or does it mean it is wrong about quantities it captures?

The second — and more fundamental — challenge is that calculating rate and Composition effects independently and then summing them fails to account for their interaction. When both rate and Composition change simultaneously between elections, the total effect is not simply the sum of the two independent effects. To see why, consider a single group: its vote outcome is the product of its margin rate $r(t)$ and its electoral share $c(t)$, giving $z(t) = r(t)c(t)$. Because this relationship is multiplicative rather than additive, changes in rate and composition interact: they cannot be cleanly separated by fixing one while varying the other. The LDA calculates the rate effect by fixing composition at t_1 and allowing rate to shift: $[r(t_2) - r(t_1)]c(t_1)$. It then calculates the composition effect by fixing rate at t_1 and allowing composition to shift: $r(t_1)[c(t_2) - c(t_1)]$. These two terms are then summed to estimate total change — but as the next equation shows, this sum is missing something.

weird to capitalize only one

Does this follow? It seems like a possibility but not a necessity

The actual total change is $\Delta z(t) = r(t_2)c(t_2) - r(t_1)c(t_1)$. Expanding this algebraically reveals the missing term:

$$\Delta z(t) = z(t_2) - z(t_1)$$

$$
\begin{aligned}
&= r(t_2)c(t_2) - r(t_1)c(t_1) \\
&= [r(t_2) - r(t_1)]c(t_1) + r(t_1)[c(t_2) - c(t_1)] + [r(t_2) - r(t_1)][c(t_2) - c(t_1)] \qquad (2)
\end{aligned}
$$

The first two terms match the LDA calculation, but the third term—$[r(t_2) - r(t_1)][c(t_2) - c(t_1)]$—represents an interaction effect. This interaction term captures the additional votes generated (or lost) when both rate and composition change together. Substantively, this interaction represents a critical dynamic: when a group both grows as a share of the electorate and shifts its rate, the impact is multiplicative rather than simply additive. For instance, if Hispanic voters both increase as a share of the electorate and shift toward Republicans, the net effect is larger than simply adding the two effects independently would suggest.

To my mind, the entire paper hinges on this. And I still don't 100% follow. I mean, I see the math. But I still think you can do a better job provably the intuition. Alternatively

When the LDA omits this interaction term, the residual must go somewhere. Omitting it implicitly assumes the interaction is zero or trivially small — but as Section 4 demonstrates, it can represent millions of votes in real elections. In the case of Marble et al. (2024)'s analysis of 2012 to 2016, this produces a 1.2 million vote residual. The problem compounds when Volume enters the picture. With three components — Rate, Composition, and Volume — the complete decomposition requires not just three main effects but also three two-way interactions and one three-way interaction:

$$
\begin{aligned}
\Delta z(t) &= z(t_2) - z(t_1) \\
&= r(t_2)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1) \\
&= [r(t_2) - r(t_1)]c(t_1)v(t_1) + r(t_1)[c(t_2) - c(t_1)]v(t_1) + r(t_1)c(t_1)[v(t_2) - v(t_1)] \qquad (3) \\
&+ [r(t_2) - r(t_1)][c(t_2) - c(t_1)]v(t_1) + [r(t_2) - r(t_1)]c(t_1)[v(t_2) - v(t_1)] + r(t_1)[c(t_2) - c(t_1)][v(t_2) - v(t_1)] \\
&+ [r(t_2) - r(t_1)][c(t_2) - c(t_1)][v(t_2) - v(t_1)]
\end{aligned}
$$

use these sentence constructions sparingly

The equation's first line contains the three main effects — what the LDA attempts to calculate. The second and third lines contain the interaction terms: the products of simultaneous changes across components. Prior work has not directly calculated Volume as a separate component, implicitly treating any unaccounted votes as a Volume

effect rather than isolating it explicitly. → Again, be super clear if this is a problem with calculating percentages or just vote totals.

As a result, methods that rely only on calculating the individual component effects do not account for how these components interact when they change simultaneously. This generates error because the LDA specification is, in essence, treating the problem as if changes in each component can be evaluated independently, when in fact the underlying relationship is multiplicative. If rate and composition changes are small, or if they move in offsetting directions, the interaction terms may be relatively minor and the LDA provides a reasonable approximation. However, as the magnitude of changes grows—or when rate and composition shifts reinforce each other—the interaction terms become substantial, and the LDA approximation deteriorates. The challenge is illustrated in Figure 1, which shows how approximations that ignore interaction effects accumulate error as the changes being measured become larger.

This is potentially really cool but needs more explication

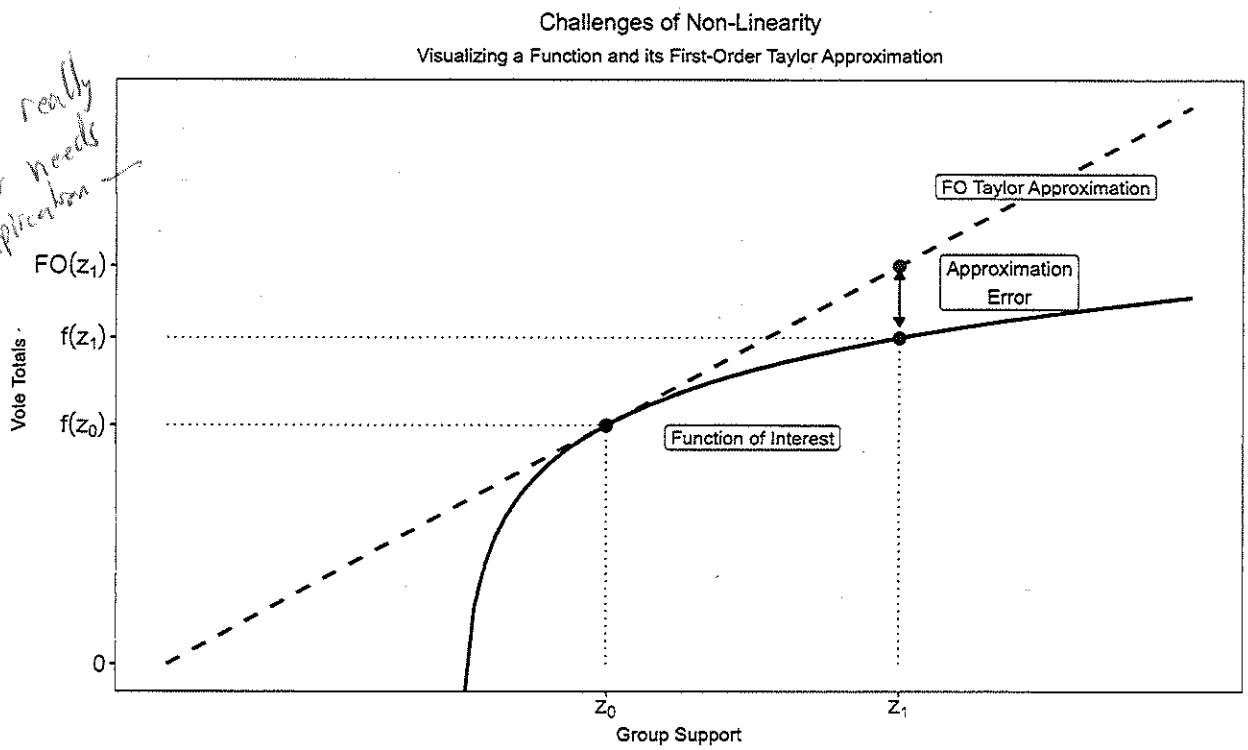


Figure 1: This figure illustrates how methods that treat changes as independent (ignoring interaction effects) accumulate error as the magnitude of change increases. The figure is based on Figure 4.3.1 in Reed (1998), showing that linear approximations become less accurate when the underlying function involves multiplicative relationships between components.

RCVD resolves the interaction term problem through sequential rather than simul-

taneous calculation. The underlying intuition is straightforward: when both rate and composition change between elections, we cannot meaningfully measure the impact of composition changes while holding rate fixed at its prior values. As Citrin et al. (2003) demonstrate, the composition of the voter pool — who actually turns out — can shift substantially across elections. Measuring composition effects using outdated rates — that is, asking what the new voter pool would have produced if the old voting patterns had held — introduces bias that is not merely a rounding error: it can reverse the estimated sign of a composition effect when a candidate crosses from winning to losing a group between elections.

Define $z_i(t)$ as the net margin votes that group i contributes to a candidate, where t indexes elections and i indexes demographic groups. Following Marble et al. (2024), the analysis works with vote margins rather than raw vote totals. Each group's contribution is the product of three quantities:

$$z_i(t) = r_i(t)c_i(t)v(t) \tag{4}$$

Where $r_i(t)$ is the margin rate at which group i votes for the candidate, $c_i(t)$ is group i's share of the total electorate, and $v(t)$ is the total number of voters. Note that $r_i(t)$ and $c_i(t)$ vary across groups while $v(t)$ is constant — the total electorate size is the same for all groups.

This specification differs from Marble et al. (2024) and Fraga, Velez, et al. (2024) in its explicit treatment of volume as a separate component. Because volume is calculated last in the sequential decomposition, it functions as a pure scaling factor: it captures only the proportional expansion or contraction of the electorate after rate and composition have already been updated to their new values. Isolating volume explicitly matters for two reasons. First, total votes provides the only firm external benchmark for validating survey-based estimates — since group compositions and vote choice are estimated from surveys, the final reported vote tally is the only hard number available

for checking plausibility, a point Section 4 demonstrates directly. Second, as Marble et al. (2024) note, this specification allows researchers to apply exit poll results to vote tallies almost immediately after an election, before detailed demographic breakdowns are available.

I propose the following sequential decomposition of $\Delta z_i(t)$ into its rate, composition, and volume components:

$$\begin{aligned}\Delta z_i(t) =& [r_i(t_2) - r_i(t_1)]c_i(t_1)v(t_1) \\ &+ r_i(t_2)[c_i(t_2) - c_i(t_1)]v(t_1) \\ &+ r_i(t_2)c_i(t_2)[v(t_2) - v(t_1)]\end{aligned} \tag{5}$$

Is this sequency a consequential assumption? Could you change the order. What are the substantive implications? What if someone who would have changed sides doesn't vote in the 2nd election?

The first term isolates the rate effect for group i: the change in $z_i(t)$ when rate shifts from $r_i(t_1)$ to $r_i(t_2)$, holding composition and volume at their initial values. The second term isolates the composition effect: the change in $z_i(t)$ when group i's electoral share shifts from $c_i(t_1)$ to $c_i(t_2)$, using the updated rate $r_i(t_2)$ but holding volume fixed. The third term isolates the volume effect: the change in $z_i(t)$ when overall turnout shifts from $v(t_1)$ to $v(t_2)$, using both the updated rate and updated composition for group i. The proof that these three terms sum exactly to $z_i(t_2) - z_i(t_1)$ — with zero residual — is in the Appendix. This is the RCVD specification.

Two features of this specification warrant attention. First, turnout affects both $c_i(t)$ and $v(t)$, while population growth affects $v(t)$ directly and $c_i(t)$ only when it is differential across groups. This raises the question of whether these should be decomposed further. They need not be, for two reasons. Across election cycles just four years apart, underlying changes in the racial composition of the potential electorate are unlikely to be substantively meaningful compared to turnout shifts. More fundamentally, turnout changes only matter insofar as they are differential across groups. If all groups increase turnout at the same rate, relative group sizes are preserved and there is no compositional effect. Universal changes — overall population growth, uniform turnout

Maybe worth a footnote w/ some stats?

increases — are absorbed into volume. Only differential changes shift composition.

A second crucial feature is that this sequence is not unique: the decomposition can be performed in any order of rate, composition, and volume. Different orderings yield different component estimates because each ordering implicitly defines a different counterfactual comparison. This is a deeper point than it might appear — every decomposition approach embeds assumptions about ordering. The LDA simply hides this by implicitly treating all cross-partial terms as trivially small. RCVD makes that assumption explicit and then eliminates the need for it entirely. The rate → composition → volume ordering is the most interpretable for two reasons. First, calculating rate effects on the first period's composition and volume ensures continuity with existing literature — the rate calculation matches prior work exactly, correcting its downstream errors without redefining them. Second, calculating volume last means it functions as a pure scaling factor: it captures only the proportional expansion or contraction of the electorate after rate and composition have already been updated to their new values. Section 3 demonstrates how alternative orderings change the interpretation of each component.

Explain substantively what this means using a voter as an example

3 Proof of Concept: Validation through Controlled Simulation

This section moves from theoretical framework to empirical application. The previous section introduced the mathematical structure of RCVD and explained why sequential calculation resolves the interaction term problem that affects the Linear Difference Approach. Here, I demonstrate how these approaches perform in practice. I begin with a simple worked example using hypothetical data to illustrate exactly how the LDA generates a residual while RCVD achieves zero error. I then apply both methods to increasingly complex scenarios: first, a two-group analysis of the 2016-2020 election showing the magnitude of the LDA's residual in real data; second, a controlled scenario isolating the rate-composition interaction by holding volume constant; and finally, a justification for the rate → composition → volume calculation sequence based on

theoretical interpretability.

A Simple Worked Example

To see precisely how the LDA generates a residual while RCVD does not, consider a straightforward hypothetical scenario with two voter groups across two elections. Following Marble et al. (2024), we work with vote margins—the difference between votes for Candidate X and votes for Candidate Y within each group. Table 1 presents the data, using round numbers throughout to allow readers to verify each calculation step.

Table 1: Hypothetical Two-Group Election Data

Group	Votes for X	Votes for Y	Total	X Margin	X Margin Rate	% of Total
Panel A: Election 1						
Group A	18	12	30	+6	+20%	30.0%
Group B	28	42	70	-14	-20%	70.0%
Total	**46**	**54**	**100**	**-8**	**-8%**	**100%**
Panel B: Election 2						
Group A	18	22	40	-4	-10%	33.3%
Group B	36	44	80	-8	-10%	66.7%
Total	**54**	**66**	**120**	**-12**	**-10%**	**100%**

Note: Margin = Votes for X − Votes for Y. Margin Rate = Margin ÷ Total Voters.

In Election 1, Group A favors Candidate X (18 votes for X versus 12 for Y, yielding a +6 vote margin and a +20% margin rate), while Group B favors Candidate Y (28 votes for X versus 42 for Y, yielding a -14 vote margin and a -20% margin rate). The total margin across both groups is -8 votes, meaning Candidate Y leads by 8 votes overall. By Election 2, both groups shift toward Candidate Y. Group A now opposes X (18 votes for X versus 22 for Y, yielding a -4 vote margin and a -10% margin rate), and Group B continues opposing X but less strongly (36 votes for X versus 44 for Y, yielding an -8 vote margin and a -10% margin rate). The total margin is now -12 votes, meaning Candidate X's margin declined by 4 votes between the two elections. This change resulted from three factors: shifts in margin rates within groups (Rate),

changes in group sizes (composition), and an increase in overall turnout (volume). Table 2 shows how the LDA and RCVD decompose this 4-vote margin change.

Table 2: LDA vs RCVD Decomposition of 4-Vote Margin Change

Component	LDA Calculation	RCVD Calculation
Rate Effect:		
Group A	$(-0.30) \times 0.30 \times 100 = -9.0$	$(-0.30) \times 0.30 \times 100 = -9.0$
Group B	$(+0.10) \times 0.70 \times 100 = +7.0$	$(+0.10) \times 0.70 \times 100 = +7.0$
Total	**-2.0 votes**	**-2.0 votes**
Composition Effect:		
Group A	$(+0.20) \times (+0.033) \times 100 = +0.667$	$(-0.10) \times (+0.033) \times 100 = -0.333$
Group B	$(-0.20) \times (-0.033) \times 100 = +0.667$	$(-0.10) \times (-0.033) \times 100 = +0.333$
Total	**+1.333 votes**	**0.0 votes**
Volume Effect:		
Group A	$(+0.20) \times 0.30 \times 20 = +1.2$	$(-0.10) \times 0.333 \times 20 = -0.667$
Group B	$(-0.20) \times 0.70 \times 20 = -2.8$	$(-0.10) \times 0.667 \times 20 = -1.333$
Total	**-1.6 votes**	**-2.0 votes**
Sum of Components	**-2.267 votes**	**-4.0 votes**
Actual Change	**-4.0 votes**	**-4.0 votes**
Residual Error	**-1.733 votes**	**0.0 votes**

RCVD running totals: -10 votes after Rate, -10 after Composition, -12 after Volume

Both methods calculate the same rate effect (-2.0 votes), but they diverge in the composition and volume calculations. The LDA holds all components at Election 1 values when calculating each effect independently. For the composition effect, it applies Election 1's margin rates (+20% for Group A, -20% for Group B) to the compositional shift (+3.3 percentage points for Group A, -3.3 for Group B), producing a +1.333 vote effect. For the volume effect, it applies Election 1's margin rates and composition to the 20-voter increase, producing a -1.6 vote effect. When summed, these components total -2.267 votes, leaving a -1.733 vote residual.

In contrast, RCVD calculates each component sequentially, using updated values at each step. After the rate effect reduces the margin from -8 to -10 votes, the composition calculation uses the *new* margin rates (-10% for both groups) rather than the old rates. Because both groups have identical margin rates in Election 2, the compositional shift from 30%/70% to 33.3%/66.7% produces zero net effect—moving voters between

two groups that vote identically has no impact on the overall margin. The volume effect then scales from 100 to 120 voters using both the new margin rates and new composition, reducing the margin by an additional 2.0 votes (from -10 to -12). The RCVD components sum exactly to -4.0 votes with zero residual.

This simple example demonstrates why RCVD is necessary: even with straightforward data and small changes, the LDA generates residuals because it cannot properly account for how changes in rate, composition, and volume interact. In real elections with larger shifts and more groups, these residuals grow substantially, as the following sections demonstrate.

Two-Group Simulation: Quantifying the Residual

The simple example above used hypothetical data to illustrate the mechanics of the LDA's residual problem. Applying the same analysis to actual election data reveals that these residuals grow substantially larger in real-world applications. Table 3 presents a simplified analysis of the 2016 to 2020 presidential election, collapsing racial groups into White and non-White voters. The data is derived from the framework introduced by Marble et al. (2024), with vote totals rounded for clarity of presentation.[2][3]

Table 3 shows the magnitude of the change we must decompose: a total shift of 9 million votes for the Democratic candidate (from Clinton's +8 million margin to Biden's +17 million margin). This shift is characterized by changes in all three fundamental parameters. With respect to rate, there was a 7.5% decline in margin within the non-White bloc, countered by a 6.6% improvement in margin within the White bloc. In terms of composition, there was a 3.6% shift of the electorate towards non-White voters. Finally, there was a large increase in volume of 26 million additional

[2]The Democratic party is consistently set as the reference group across all analyses. Positive numbers indicate increased support/votes for Democrats; negative numbers indicate decreased support. No political endorsement should be inferred from this mathematical designation.

[3]I have replicated the process used by Marble et al. (2024) in creating this data, which is based upon the data from ANES (2021). The difference in their reported votes from reality seems to be related to their estimation of voter support for candidates based on the ANES. Vote totals are rounded to the nearest half-million for ease of presentation; Section 4 provides more precise figures.

Is it possible to give intuition about where the residuals come from?

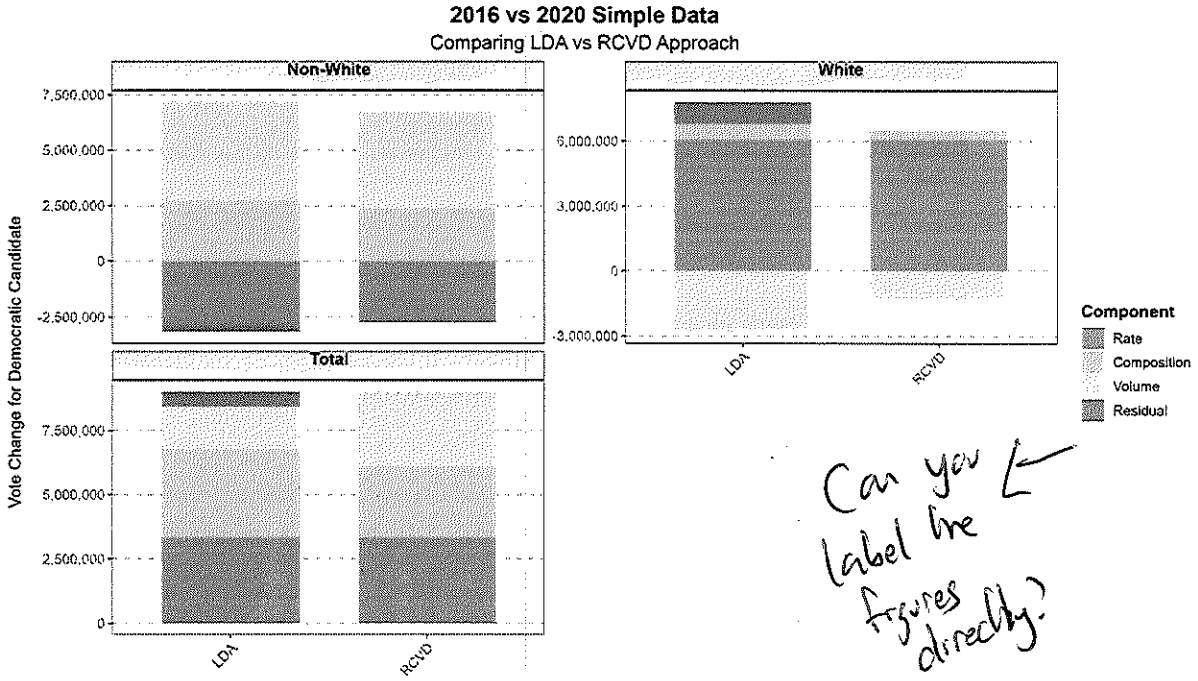
total votes cast.

Table 3: Comparing Two Elections — Basic Data

Election	Group	Candidate 1	Candidate 2	Total	Margin	Margin Rate (%)	Composition (%)
1st Election (2016)	Non-White	29.5	7.5	37.0	22.0	59.5	28.7
	White	39.0	53.0	92.0	−14.0	−15.2	71.3
	Total	68.5	60.5	129.0	8.0	6.2	100.0
2nd Election (2020)	Non-White	38.0	12.0	50.0	26.0	52.0	32.3
	White	48.0	57.0	105.0	−9.0	−8.6	67.7
	Total	86.0	69.0	155.0	17.0	11.0	100.0
Comparing the Two Elections	Non-White	8.5	4.5	13.0	4.0	−7.5	3.6
	White	9.0	4.0	13.0	5.0	6.6	−3.6
	Total	17.5	8.5	26.0	9.0	4.8	0.0

Note: Votes are reported in millions. Candidate 1 refers to Clinton (2016) and Biden (2020); Candidate 2 refers to Trump.

Figure 2 demonstrates how the LDA fails to fully account for the total 9 million vote change. When applied to this two-group system, the LDA generates a Mean Absolute Error (MAE) of just over 1.4 million votes. This unallocated portion is the residual term—real votes that contributed to the 2020 outcome but are mathematically unassigned to any specific component (Rate, composition, or volume). why is only this capitalized?



Can you label the figures directly?

Figure 2: This figure shows that even in a simple case with only two voter groups, the LDA calculation can yield misattributed votes on an order of magnitude of over one million votes. The underlying numbers reflected in this figure are in Appendix A.

This non-zero residual creates interpretive problems. Since previous decomposition

Earlier, it would have been helpful to clarify why we need to

studies have left the volume term unspecified, they have, in essence, folded the residual into an error term that also contained the correctly calculated volume effect. As demonstrated in Figure 3, this leads to systematic misattribution. The benefit Biden received from the change in the racial composition of the electorate is understated by 650,000 votes using the LDA, while a large portion of the 1.4 million residual is incorrectly assigned, masking the true contribution of the volume effect.

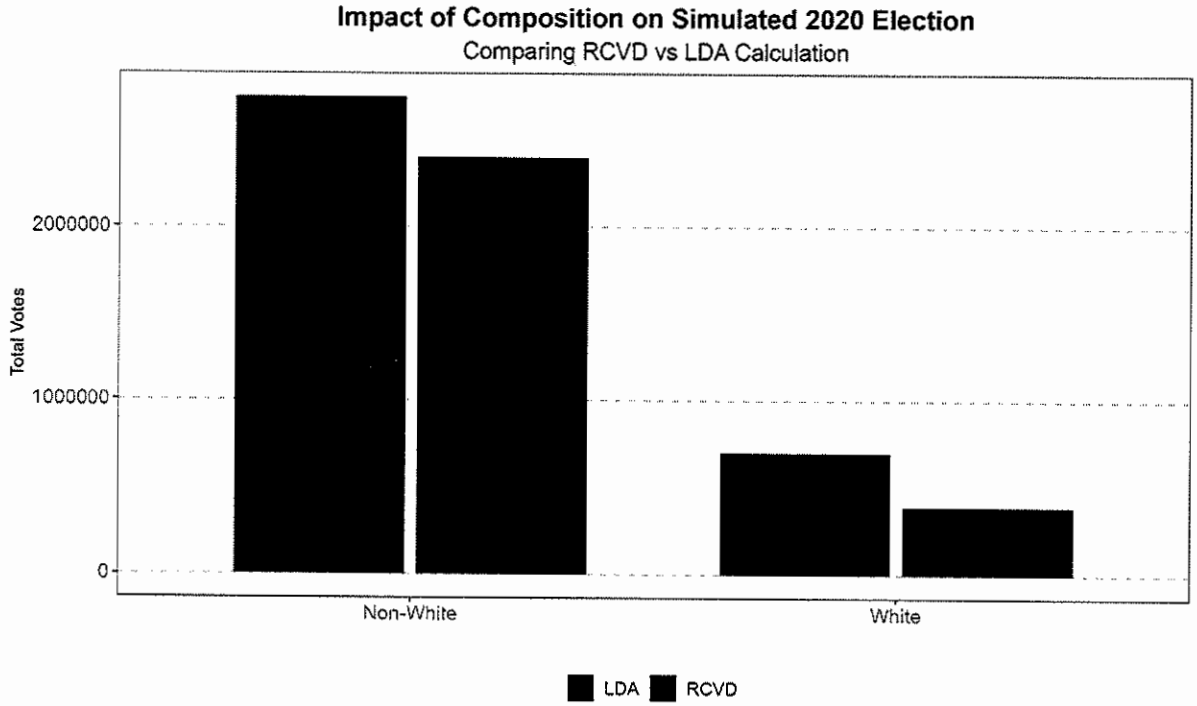


Figure 3: This figure shows that even in a simple case with only two voter groups, the LDA calculation can misattribute the impact of shifts in composition on an order of magnitude of hundreds of thousands of votes. The underlying numbers reflected in this figure are in Appendix A.

By contrast, the RCVD approach leaves a MAE of 0, precisely accounting for every vote in the system. The RCVD approach retains the highly intuitive property of explaining rate in a manner identical to the LDA, meaning that the large body of existing work focused solely on rate changes remains valid, while the problematic composition and volume terms are corrected.[4][5]

[4]See Appendix for the robustness of the method to partitions.

[5]A natural question is whether the LDA's residual might simply result from omitting the volume component

Non-Uniqueness and the Fixed RCVD Path

As noted in the conclusion of Section 2, the Rate, Composition, and Volume Decomposition (RCVD) can be calculated in any order. Mathematically, there are six possible permutations of the rate (ΔR), composition (ΔC), and volume (ΔV) terms that satisfy the zero-loss property of the RCVD framework. This path-dependence is inherent to correctly modeling the interaction effects that the LDA omits. However, different orderings yield different interpretations of each component's contribution, as each ordering implicitly defines a different counterfactual comparison. Therefore, the chosen order ($\Delta R \rightarrow \Delta C \rightarrow \Delta V$) requires substantive justification.

Is this need to identify the order inherent to any approach?

Figure 12 illustrates an alternative zero-loss order (composition $\rightarrow$ rate $\rightarrow$ volume) when applied to the simple 2016-2020 simulation. While the total MAE remains 0, the calculated contribution of rate and composition shifts: the alternative order assigns a slightly larger role to composition and a slightly smaller role to rate.

rather than from mismodeling interaction terms. The appendix addresses this directly by analyzing a controlled scenario where volume is held constant at zero. Even with no volume change, the LDA still produces a residual of 1.7 million votes, definitively demonstrating that the residual stems from the interaction term problem, not from volume omission.

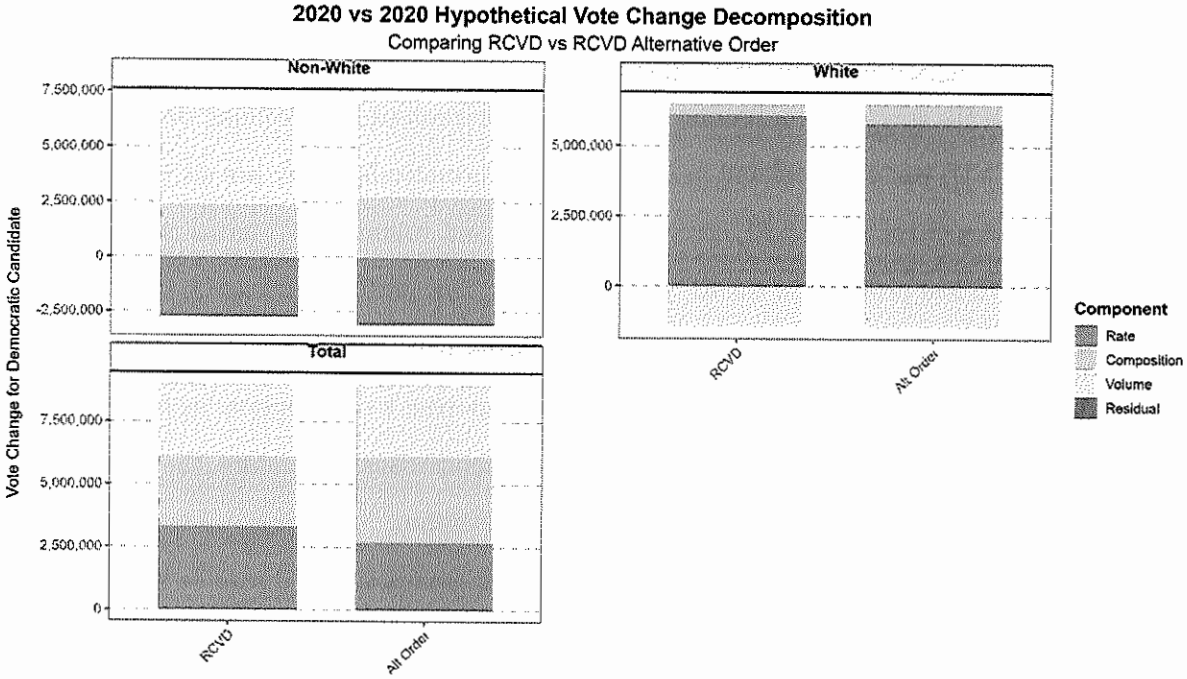


Figure 4: This figure shows that shifting the order in which RCVD is calculated yields no error term, but can shift the interpretation of the impacts of both rate and composition.

Is this assuming something about the relative timing of shifts in voter preferences vs. turnout choices? Maybe that's the way to motivate it with substantive research?

The $\Delta R \rightarrow \Delta C \rightarrow \Delta V$ ordering provides the most interpretable decomposition for two reasons. First, this order ensures continuity with existing literature. The convention in electoral studies has always been to calculate the change in group loyalty (Rate) based on the prior period's electorate composition. By sequencing rate first, the ΔR term captures preference changes holding composition and volume at their initial values. This means the rate calculation in RCVD produces exactly the same value as in the LDA, ensuring that the large body of existing work focused on rate changes remains valid while the composition and volume terms are corrected.

Second, this order ensures that volume functions as a proportional scaling factor. By sequencing volume last, we calculate it after rate and composition have already been updated. This means volume captures only the impact of expanding or contracting the electorate while preserving the relationships established by the new rate and new composition. Intuitively, if we shifted composition first (say, increasing the non-White share of the electorate) before calculating volume, the volume effect would be confounded with compositional changes. By placing volume last, it represents a clean

proportional increase or decrease in the system's size, distributed across the electoral configuration determined by rate and composition.

In sum, the $\Delta R \rightarrow \Delta C \rightarrow \Delta V$ ordering ensures that each component has a clear, theoretically meaningful interpretation that aligns with how electoral scholars conceptualize these dynamics.

4 Real-World Application: ANES, Pew, and the 2024 Hispanic Vote

The preceding section established the rate, composition, and volume Decomposition (RCVD) framework and demonstrated that the Linear Difference Approach obscures the mechanisms driving electoral change. This section applies RCVD to analyze recent U.S. presidential elections, with particular attention to the Hispanic electorate.

The analysis proceeds in three parts. First, I apply RCVD to the 2020 election using ANES data (ANES, 2021), showing how it reveals different interpretations of compositional effects compared to LDA methods. Second, I compare ANES and Pew data for the 2020 election, demonstrating that Pew provides greater descriptive accuracy for analyzing Hispanic voter behavior. This comparison supports recent findings by Fraga, Velez, et al. (2024) on Hispanic electoral volatility. Finally, I analyze three electoral cycles (2016–2020, 2020–2024, and 2016–2024) using Pew data, isolating how shifts in voter preferences (Rate), demographic composition (composition), and overall turnout (volume) shaped the 2024 election outcome.

Why these cases?

Doesn't it need to be two elections?

The 2020 Baseline: Compositional Effects in the ANES

Political analysis of the 2020 presidential election typically relies on changes in two-party vote margins within demographic groups.[6] However, a change in a group's aggregate vote margin may result from a shift in partisan preference (Rate), a change in that group's share of the electorate (composition), or overall turnout changes (volume).

[6]Following Marble et al. (2024), I work with two-party vote margins throughout this analysis. A margin represents the difference in vote shares between parties: for example, if Democrats receive 72% and Republicans 28% of a group's votes, the Democratic margin is +44 percentage points. While I report some results in terms of vote shares for interpretability, the RCVD framework decomposes margin changes.

Standard decomposition methods fail to separate these effects cleanly, hindering our understanding of what drove the election outcome.

I begin by applying the RCVD framework to data from the 2020 American National Election Studies (ANES) (ANES, 2021), which represents the discipline's historical "gold standard." While ANES has limitations in describing recent election outcomes (discussed in the next subsection), it provides a useful starting point for demonstrating RCVD's value.

citations

Conventional analysis suggests that the Democratic victory was driven by a modest rate shift among non-White voters combined with significant compositional advantages. The RCVD decomposition reveals a more nuanced picture. By correctly partitioning the total vote shift, I find that compositional effects are significantly less beneficial to Democrats than LDA analyses suggest. Figure 5 shows how the LDA overstates the positive impact of compositional shifts for Democrats by roughly 365,000 votes.

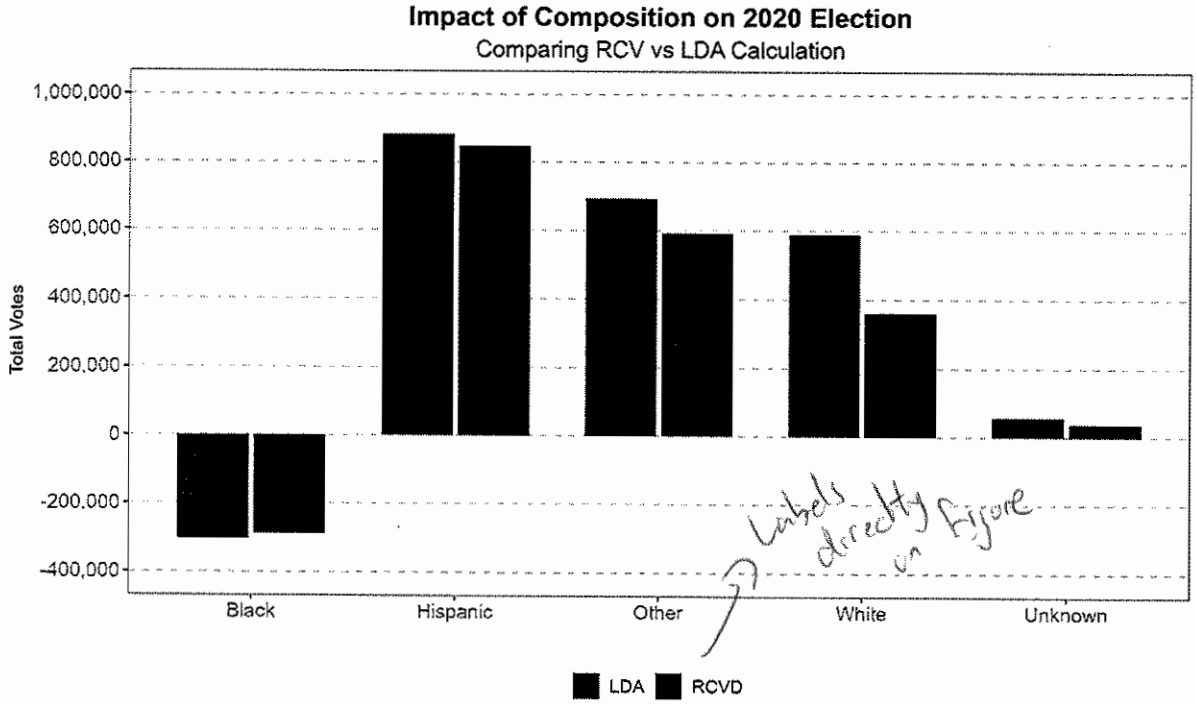


Figure 5: This figure shows how shifting from the LDA calculation to the RCVD calculation changes the evaluation of the impact of composition on the 2020 election. In particular, the LDA calculation overemphasizes the benefits of composition towards Democrats by several hundred thousand votes.

The RCVD framework provides a zero-loss allocation, ensuring that the sum of the rate, composition, and volume components precisely equals the observed overall change in the national vote margin. This contrasts with LDA methods that leave unexplained residual terms. Figure 6 shows that for White voters, the LDA's error actually exceeds the total impact of composition on the final outcome. The total error across all groups is roughly 700,000 votes versus a total compositional impact (from RCVD) of roughly 1.5 million votes.

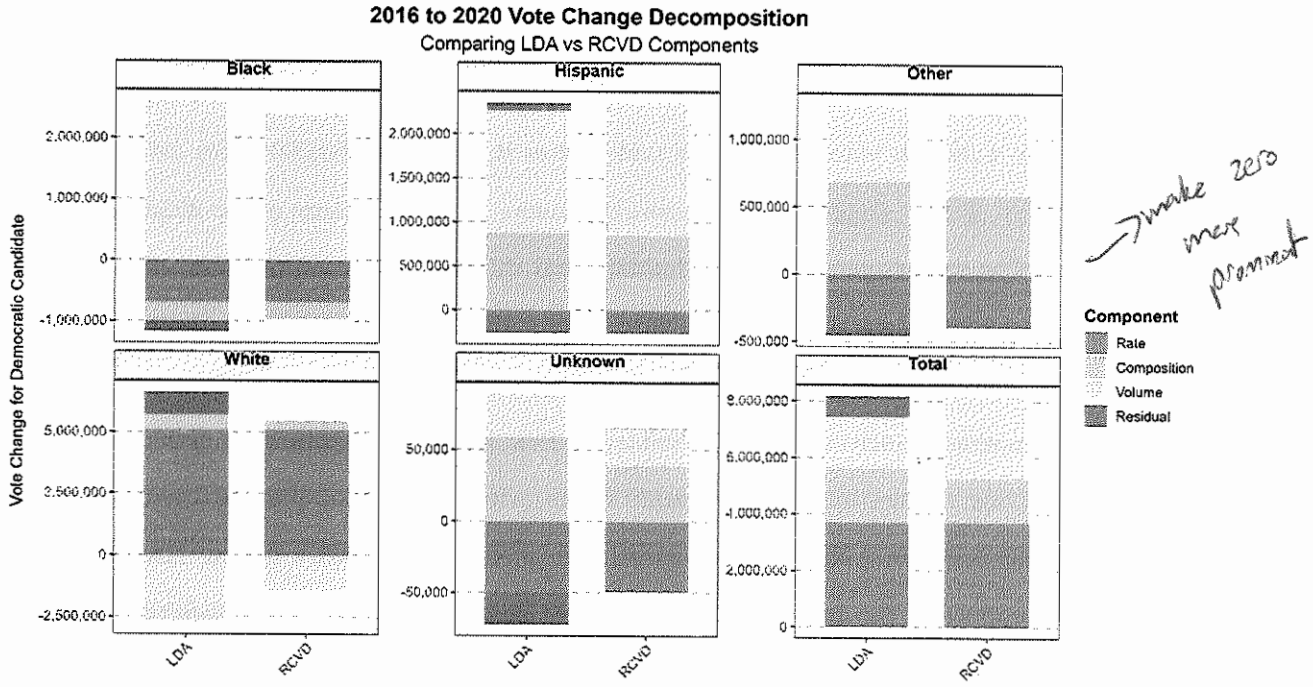


Figure 6: This figure shows how shifting to RCVD shifts interpretations of the 2020 election results when compared to 2016. It highlights that the magnitude of the error associated with the LDA is oftentimes several hundred thousand votes, and in the case of White voters, actually exceeds the total impact of composition.

The application of RCVD to the 2020 ANES data demonstrates that standard methods risk substantial bias in understanding how composition affects election outcomes. This finding motivates the subsequent analysis, where I show that Pew data provides greater accuracy in describing recent elections and use it to analyze the increasingly salient Hispanic electorate.

Survey Divergence: Comparing ANES and Pew

The robustness of any electoral decomposition depends on the descriptive accuracy of the underlying survey data. While ANES remains the discipline's preferred benchmark, its ability to capture rapidly evolving preferences among the Hispanic electorate has been questioned (Fraga, 2016). I therefore compare ANES to Pew Research Center's validated voter data (Hartig et al., 2023), which employs a different sampling strategy and may achieve better resolution for non-White subpopulations.

Figure 7 compares the two datasets' accuracy and their RCVD decompositions for Hispanic voters. The lower panel shows that ANES overstates Democratic support by nearly 4 million votes, while Pew understates it by under 600,000 votes (Wooley and Peters, 2025)—suggesting Pew provides a more accurate picture of the 2020 electorate. The upper panel reveals a fundamental difference in interpretation. ANES shows a modest Republican gain in rate among Hispanic voters but attributes the overall Democratic increase from this group to composition— a potentially persistent, demographic-driven effect. In contrast, Pew isolates a significantly larger rate effect driving the net impact of Hispanic voters, with composition playing a minimal role. This suggests ANES may underestimate preference volatility among Hispanic voters, supporting the argument in Fraga, Velez, et al. (2024) about the magnitude of Hispanic electoral shifts.

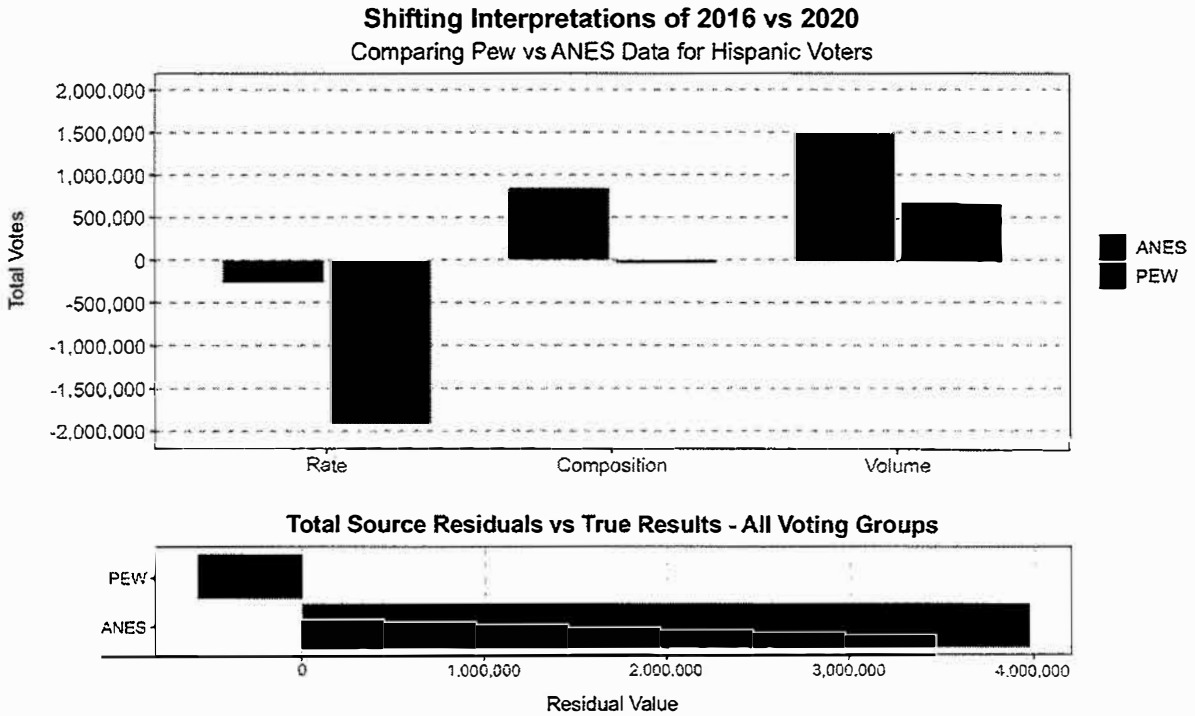


Figure 7: This figure compares the implied election results from ANES data versus Pew data. The upper portion of the figure shows that utilizing Pew data fundamentally alters the interpretation of the impact of Hispanic voters on the 2020 election. In particular, it shows that there is no evidence that Democrats benefited from a shift into Hispanic voters in 2020 and instead suffered a significant rate decline. The lower portion of the figure shows that the implied error in votes of using ANES data to evaluate the 2020 election is nearly 4 million votes, whereas using the Pew data has an error of less than 600,000.

Aren't there better cites on this?

This difference in interpretation carries strategic implications. If the change is mostly compositional (as ANES suggests), the Democratic challenge is primarily organizational. If the change is driven by rate effects (as Pew suggests), the challenge is fundamentally one of persuasion. The Pew data thus refutes the notion that Democrats can rely on demographic changes alone (Klein, 2024), revealing deeper vulnerability to Republican outreach than previously recognized.

This comparison serves as a methodological caution: researchers should carefully assess whether their data reflects reality, as argued in Grimmer, Marble, et al. (2022). By embracing the superior descriptive validity of Pew for the Hispanic electorate, the RCVD framework establishes that Hispanic volatility is driven less by passive demographic turnover and more by active shifts in partisan loyalty—a crucial baseline

for understanding the dramatic shifts in the 2024 election.

The Full Picture: Decomposition Across Three Electoral Cycles

I now apply the RCVD framework to analyze vote change across three electoral cycles: 2016–2020, 2020–2024, and the full 2016–2024 interval. This analysis demonstrates how the compounding effects of rate and composition determined the 2024 electoral outcome.

The 2016 to 2020 Cycle— Figure 8

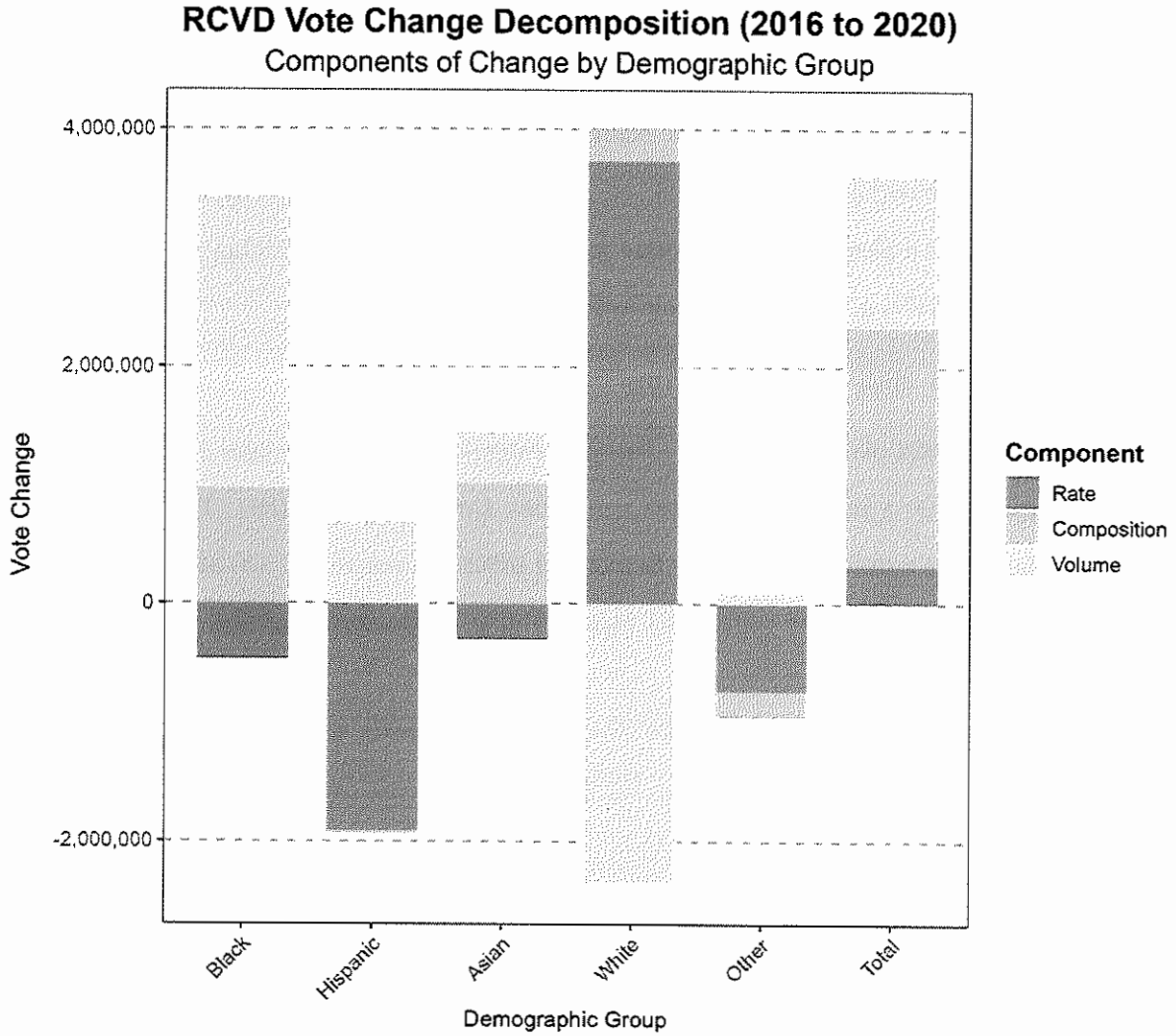


Figure 8: This figure provides an evaluation of the 2020 election compared to the 2016 election using Pew data and the RCVD approach. It shows that composition had a large positive impact for Democrats as a result of shifting out of White voters and into Black and Asian voters. There was also a large positive impact for Democrats driven by improved margins among White voters. However, this impact was almost entirely offset by declining Democratic margins among Hispanic voters, and to a lesser extent among Black, Asian and other voters.

The 2016 to 2020 period represents the Democratic bounce-back from their 2016 loss. While accounting for the translation of the popular vote to the Electoral College involves challenges (Grimmer, Knox, et al., 2024), Democrats clearly performed better in 2020. The victory was characterized by favorable compositional forces (increasing

share of non-White voters, particularly Black and Asian voters) combined with positive rate effects among White voters. The RCVD decomposition confirms that mobilization efforts and demographic trends worked synergistically with preference shifts to yield a substantial Democratic gain. This period represents the high-water mark of the conventional wisdom that demographic change inherently favors Democrats. However, even in 2020, warning signs appeared: all non-White groups voted for Democrats at lower rates, with Democratic vote share among Hispanic voters declining noticeably (from 72% to 61%, a margin shift from +44% to +22%).

The Critical 2020 to 2024 Swing— Figure 9

The 2020 to 2024 period represents a critical realignment that challenges established assumptions about the stability of the non-White Democratic coalition. The RCVD analysis reveals a dramatic reversal: Democrats suffered a severe net loss attributable overwhelmingly to the rate component across all non-White voting groups.

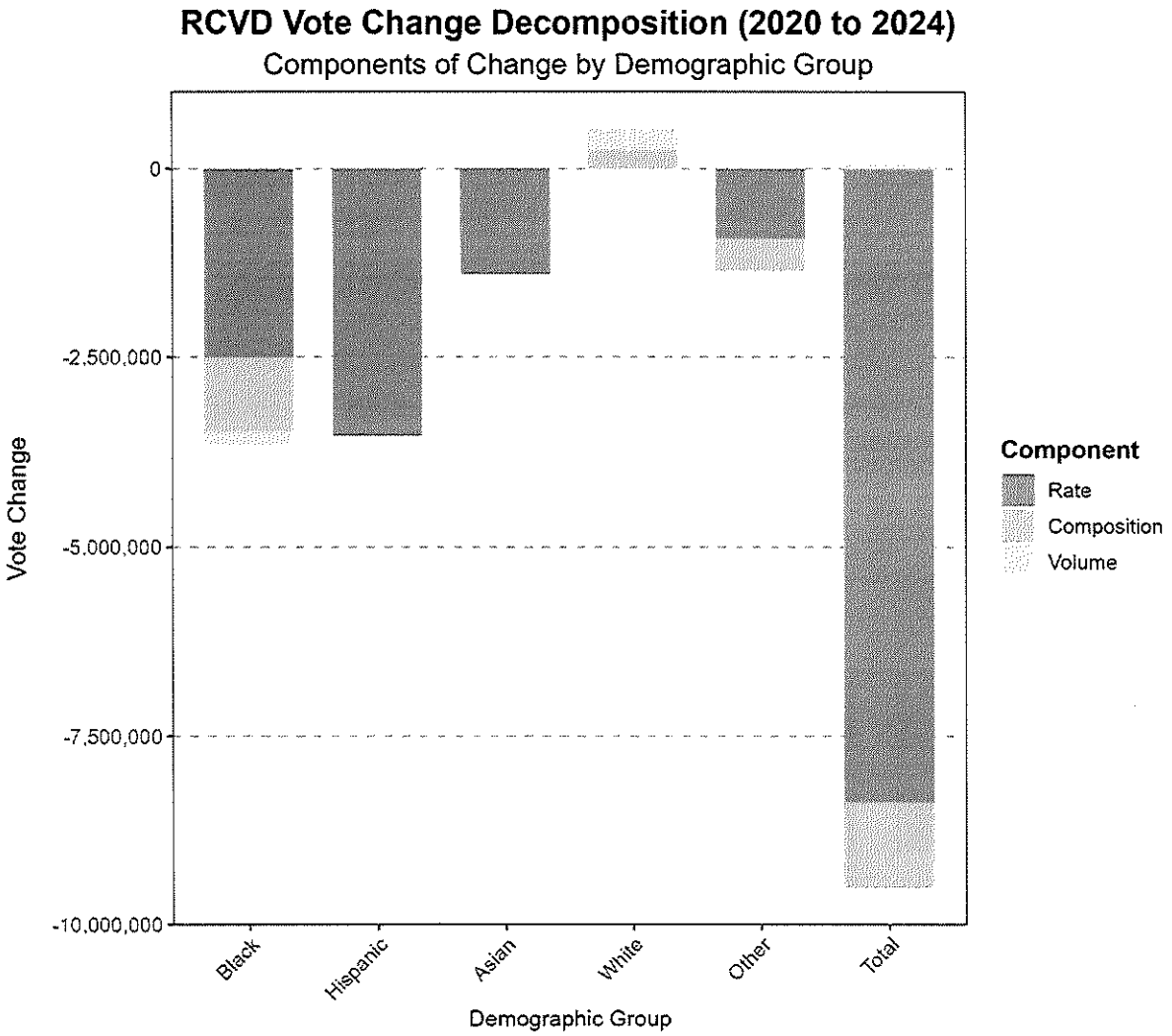


Figure 9: This figure provides an analysis of the 2024 election compared to the 2020 election. It shows that Democrats suffered a dramatic decrease in vote margins among all non-White voting groups, most prominently among Hispanic voters. The decline in support was so significant that even with continued shifts away from White voters amongst the composition of the electorate, the total impact of composition in 2024 was a net negative for Democrats.

The rate effect was negative and substantial among Hispanic and Black voters, signaling erosion of Democratic loyalty and successful Republican persuasion. This negative rate swing was so significant that, even with continued shifts away from White voters in the electorate's composition, the total composition effect in 2024 was net negative for Democrats. This finding underscores the core mechanism of the 2024 result: declining base loyalty (Rate) entirely neutralized and then overcame the structural advantages of demographic change (composition).

The Cumulative 2016 to 2024 Arc—Figure 10

The eight-year view from 2016 to 2024 provides the final synthesis. While the analysis demonstrates some favorable impacts from composition over the long run, this benefit was dwarfed by the cumulative negative trend in the rate component.

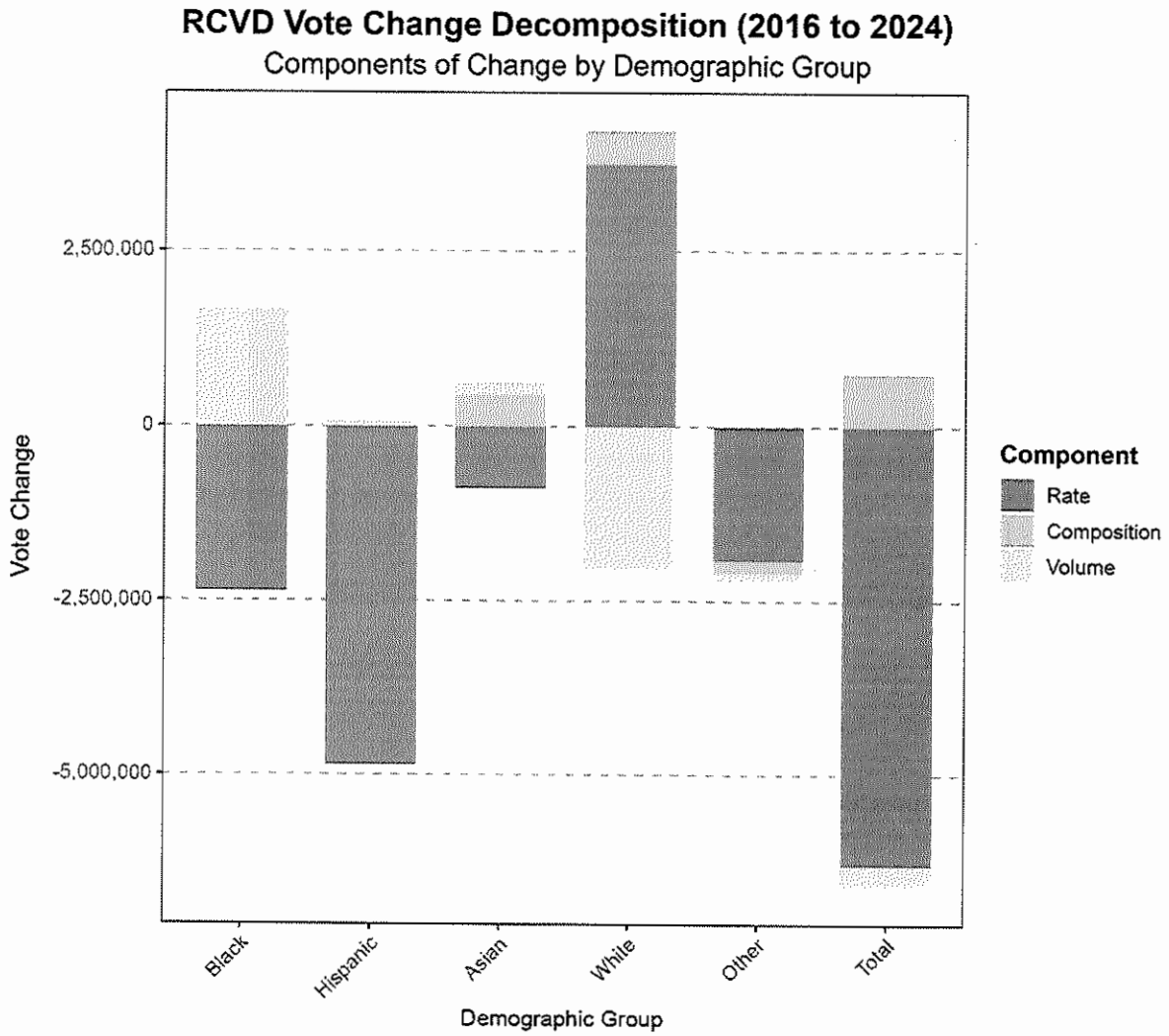


Figure 10: This figure compares the 2024 election to the 2016 election. It highlights the strong negative trend in Democratic vote margins among Hispanic voters, and to a lesser extent among other non-White voters. While this analysis demonstrates some favorable impacts from composition, it highlights the shortcomings of a reliance on these demographic shifts to promote future success for the Democratic party.

The negative rate trend among Hispanic voters is the single most important factor identified by the RCVD framework over this period. While Democrats showed positive

growth among White voters, the framework cleanly partitions the positive effect of an increasing non-White share from the negative effect of these groups becoming less reliably Democratic. The final outcome is explained not by failure of the electorate to grow (volume) or shift demographically (composition), but by failure of the Democratic Party to retain the loyalty of the electorate that did grow (Rate).

The application of RCVD across these three cycles demonstrates that the 2024 result was driven by a decisive decoupling of composition and rate effects. The long-predicted demographic dividend (composition) was neutralized by a collapse of partisan preference (Rate) within the fastest-growing segments of the electorate, a phenomenon masked by aggregate analysis and only revealed by the rate, composition, and Volume Decomposition.

5 Conclusion

As demographic shifts reshape the U.S. voting landscape, understanding how changes between groups affect election outcomes is crucial for both scholars and policymakers. Previous methods for explaining these shifts have introduced significant interpretive errors, frequently amounting to millions of misallocated votes, due to the conflation of preference shifts with compositional change. The Rate, Composition, and Volume Decomposition (RCVD) approach, as defined and applied in this paper, resolves this fundamental ambiguity. It provides a zero-loss methodology that precisely specifies how changes in partisan preference (Rate), the relative size of groups (composition), and overall participation (volume) influence electoral outcomes. The RCVD framework is broadly applicable, extending beyond U.S. elections to any context where subdividing populations into distinct groups helps explain dynamics in time-series cross-sectional data.

My empirical application of RCVD to the 2016, 2020, and 2024 presidential elections yields a core finding that fundamentally reconfigures the prevailing "demographics are destiny" narrative. The 2024 election was driven not by the expected long-term compositional benefit for the Democratic Party, but by a decisive decoupling of compo-

sition and rate effects. The framework isolates a substantial negative rate effect—the loss of partisan loyalty—among the fastest-growing non-White electorates, particularly Hispanic voters. This preference volatility neutralized the structural compositional advantage, confirming that electoral success depends less on passive demographic shifts and more on active defense of coalition loyalty. The RCVD framework is thus crucially diagnostic, directing attention from organizational success (volume and composition) toward persuasion challenges (Rate).

Future work will explore the robustness and granularity of the RCVD approach in two key areas. First, I will investigate its ability to account for nested subgroups, demonstrating how shifts within smaller partitions—such as regional or age-based splits within the Hispanic electorate contribute to broader electoral changes, building on the work of Fraga, Velez, et al. (2024). Second, I will extend the analysis across multiple non-presidential time periods to capture long-term trends, showing how RCVD can reveal the evolving durability and fragility of group shifts over time. These extensions will further validate RCVD as a tool for analyzing historical shifts in political support and for understanding electorate dynamics in future elections.